

	$= 6.2 \times 10^{-20} \text{ J}$ (accept 1 sig.fig.)	A1	[2]
4	(a)(i) $\omega = 2\pi f$	C1	
	$= 2\pi \times 1400$		
	$= 8800 \text{ rad s}^{-1}$	A1	[2]
	(ii) $a_0 = (-)\omega^2 x_0$	C1	
	$= (8800)^2 \times 0.080 \times 10^{-3}$		
	$= 6200 \text{ m s}^{-2}$	A1	[2]
	(b) straight line through origin with negative gradient	M1	
	end points of line correctly labelled	A1	[2]
	(c)(i) zero displacement	B1	[1]
	(ii) $v = \omega x_0$	C1	
	$= 8800 \times 0.080 \times 10^{-3}$		
	$= 0.70 \text{ m s}^{-1}$	A1	[2]

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